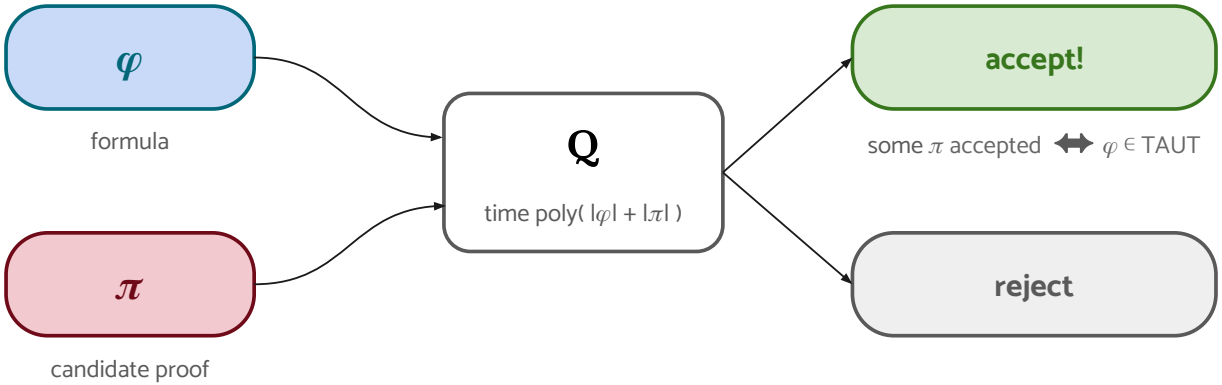
Short Propositional Proofs are Hard to Find: a Meta-Complexity Approach

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Master's Degree in Computer Science
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Academic year • 2025/2026
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What is proof complexity?



How long must π be with respect to $|\varphi| = n$?

Two examples

Resolution

Lines are propositional clauses

$$\frac{C \vee x \qquad D \vee \neg x}{C \vee D}$$ **Cut**

$$\frac{C}{C \vee x}$$ **Weakening**

e.g. $(\neg x_1 \vee x_3 \vee x_2), (\neg x_1 \vee x_4) \vdash (x_3 \vee x_2 \vee x_4)$

Polynomial Calculus

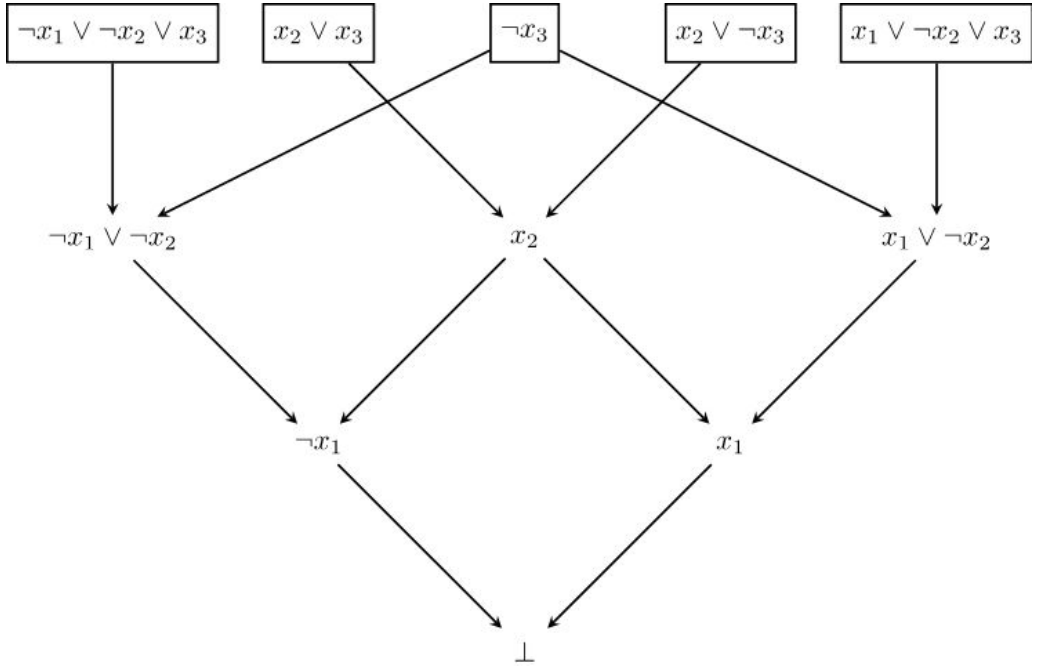
Lines are polynomials, read as $p = 0$

$$\frac{p \qquad q}{\alpha p + \beta q}$$ **Linear combination**

$$\frac{p}{x \cdot p}$$ **Multiplication**

axioms $x^2 - x = 0$ (forces $x \in \{0,1\}$)

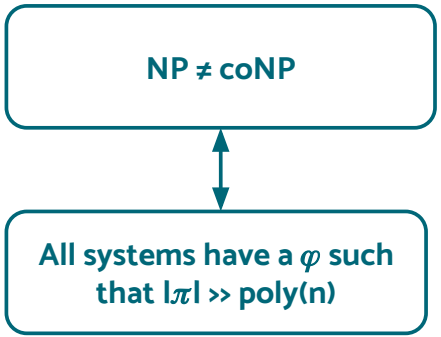
A Resolution refutation

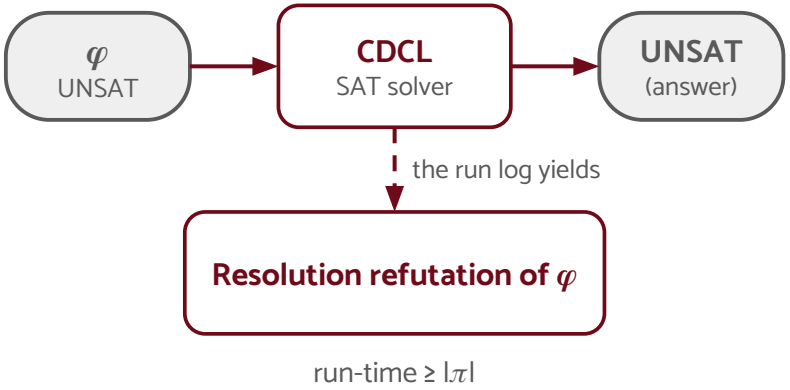


Why finding short proofs matters

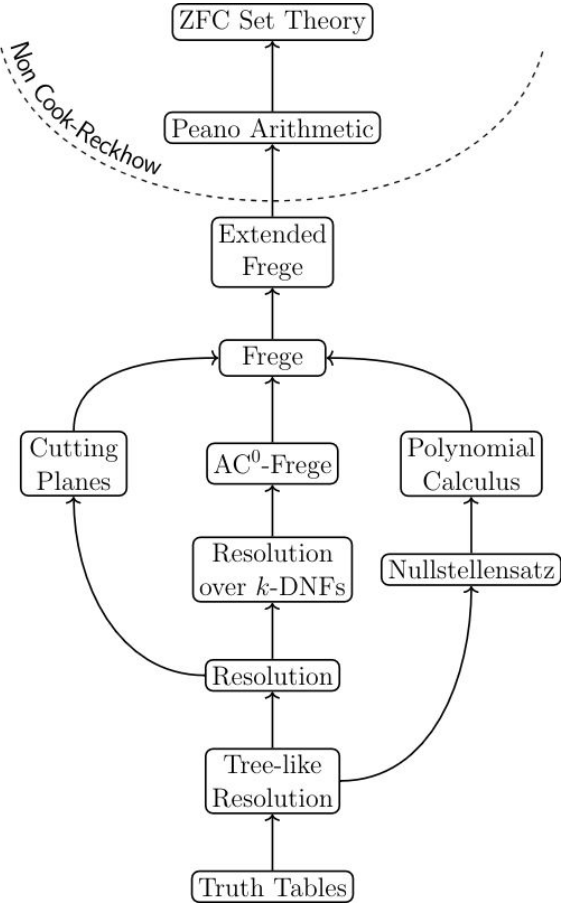
Theory: a route towards NP ≠ coNP

Practice: solvers produce proofs

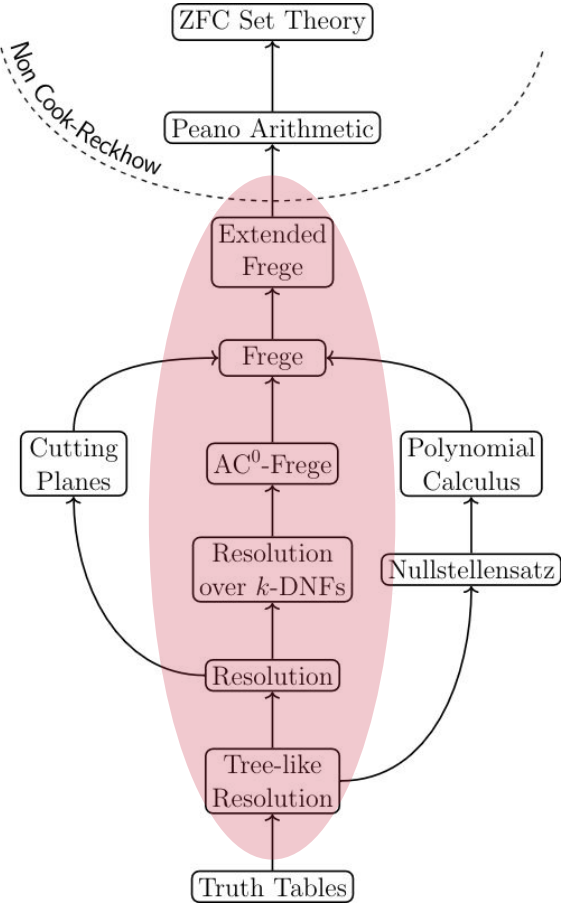




The zoo

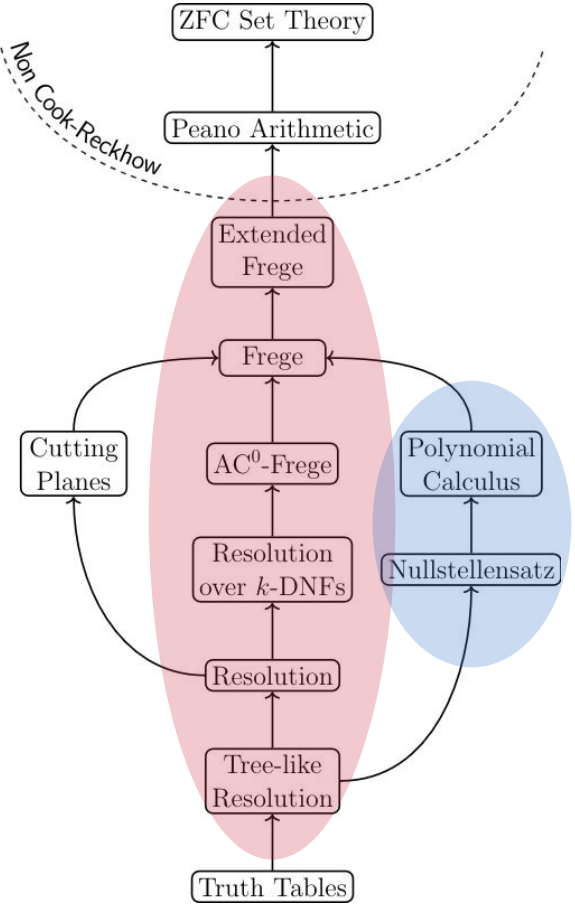

ZFC Set Theory
Non Cook-Reckhow
Peano Arithmetic
Extended Frege
Frege
Cutting Planes
AC^0-Frege
Polynomial Calculus
Resolution over k-DNFs
Nullstellensatz
Resolution
Tree-like Resolution
Truth Tables

The zoo



Logic-based systems

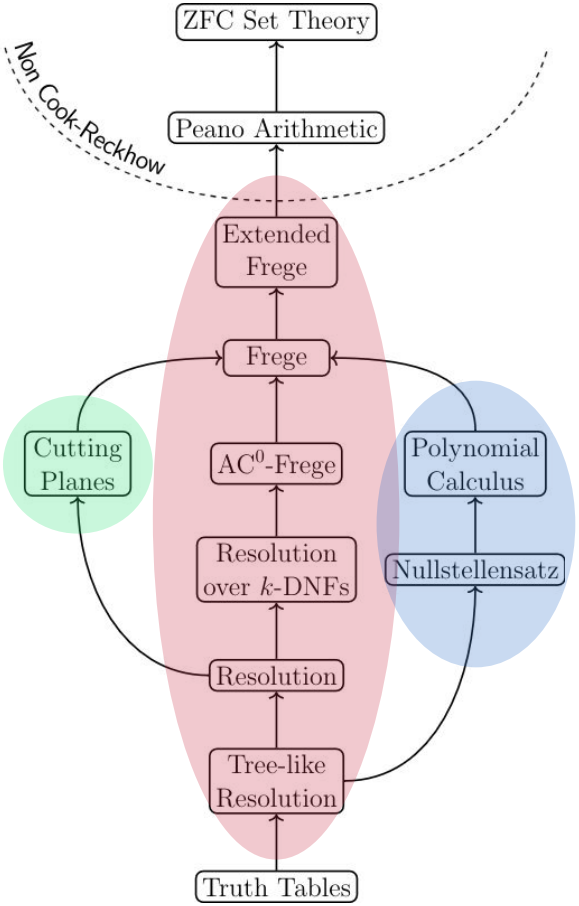
The zoo



Logic-based systems

Algebraic systems

The zoo



Logic-based systems

Algebraic systems

Geometric systems

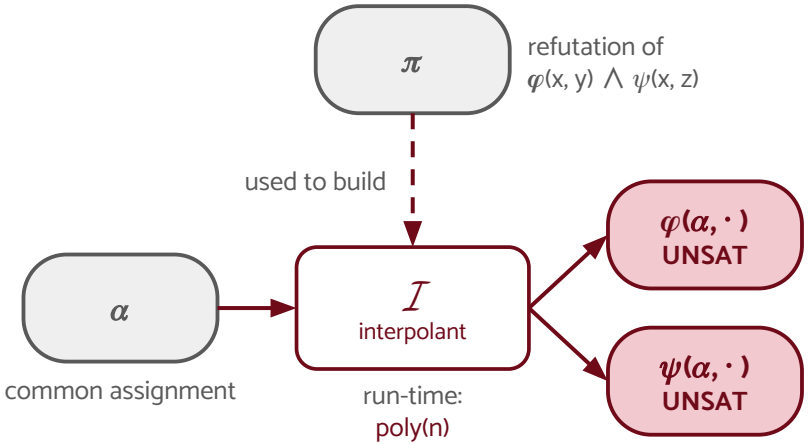
Two properties

Automatability — *Can we find short proofs for formulas?*

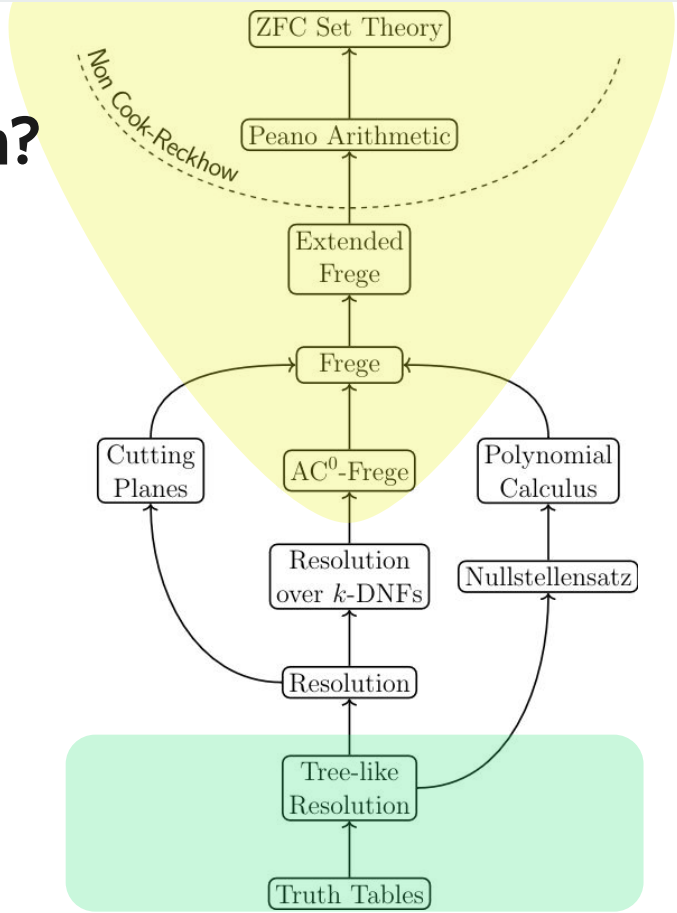
implies

Feasible interpolation — *Can we extract interpolants from proofs of split formulas?*





What is known?

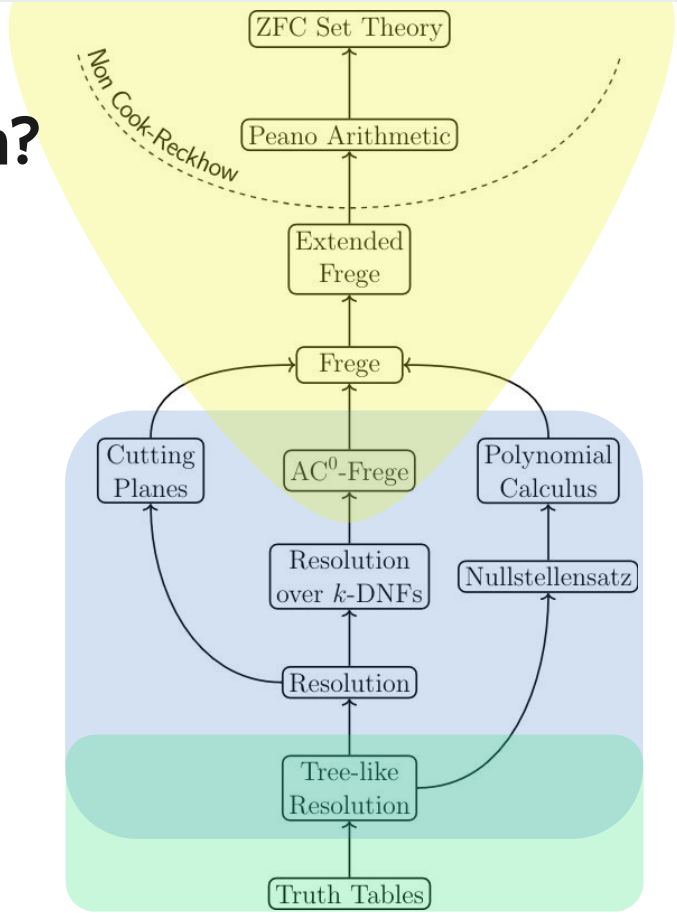


Non-automatable

No feasible interpolation unless Diffie-Hellman fails

(Quasi) Automatable!

What is known?



Non-automatable
No feasible interpolation unless Diffie-Hellman fails

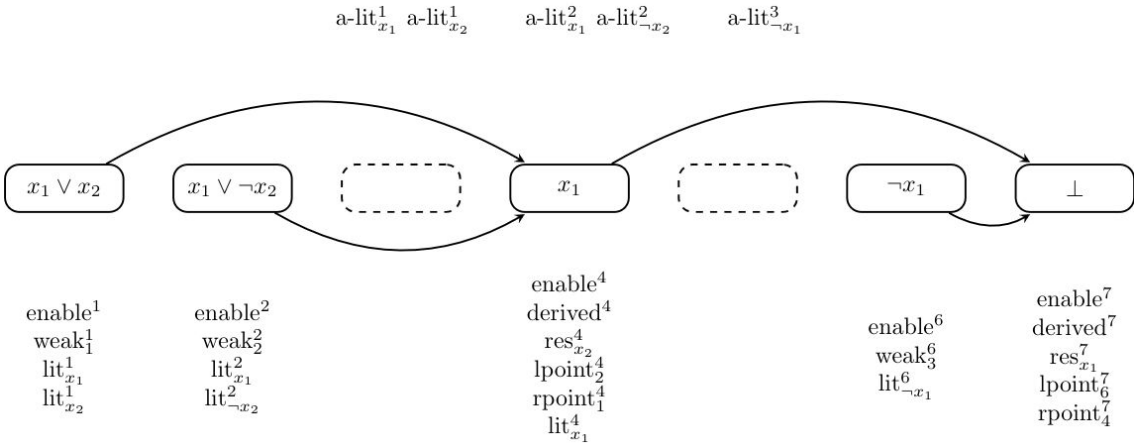
Non-automatable
Unless P = NP (or unless ETH fails)

(Quasi) Automatable!

A Meta-Complexity Approach

The Refutational Formula

$\mathbf{Ref}_s(\varphi)$ encodes the statement "φ has a Resolution refutation of size at most s"



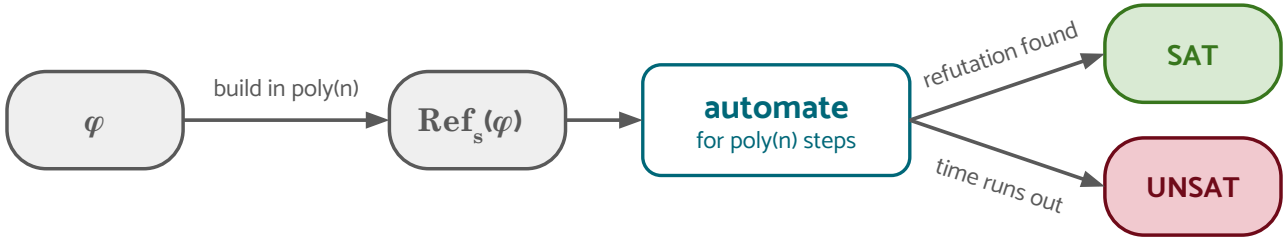
A Meta-Complexity Approach

When φ is satisfiable

$\mathbf{Ref}_s(\varphi)$ has polynomial size Res-refutations

When φ is unsatisfiable

$\mathbf{Ref}_s(\varphi)$ requires exponential size Res-refutations



New contribution

Size improvement

generalizing results for $\mathrm{Ref}_s(\varphi)$ to other values of s

before — only $s = n^2$

now — every $s \geq n^2$

simpler proofs for known results

System improvement

extending results for $\mathrm{Ref}_s(\varphi)$ to an algebraic proof system

before — in Resolution

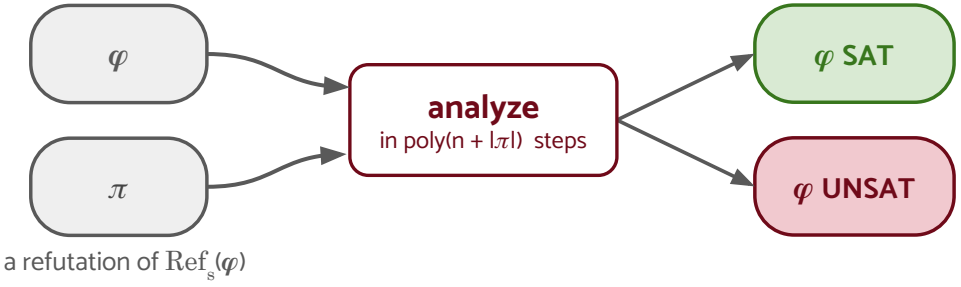
now — in Polynomial Calculus with Resolution

algebraic systems previously used variants of $\mathrm{Ref}_s(\varphi)$

one formula $\mathrm{Ref}_s(\varphi)$ • for any $s \geq n^2$ • in Resolution and PCR

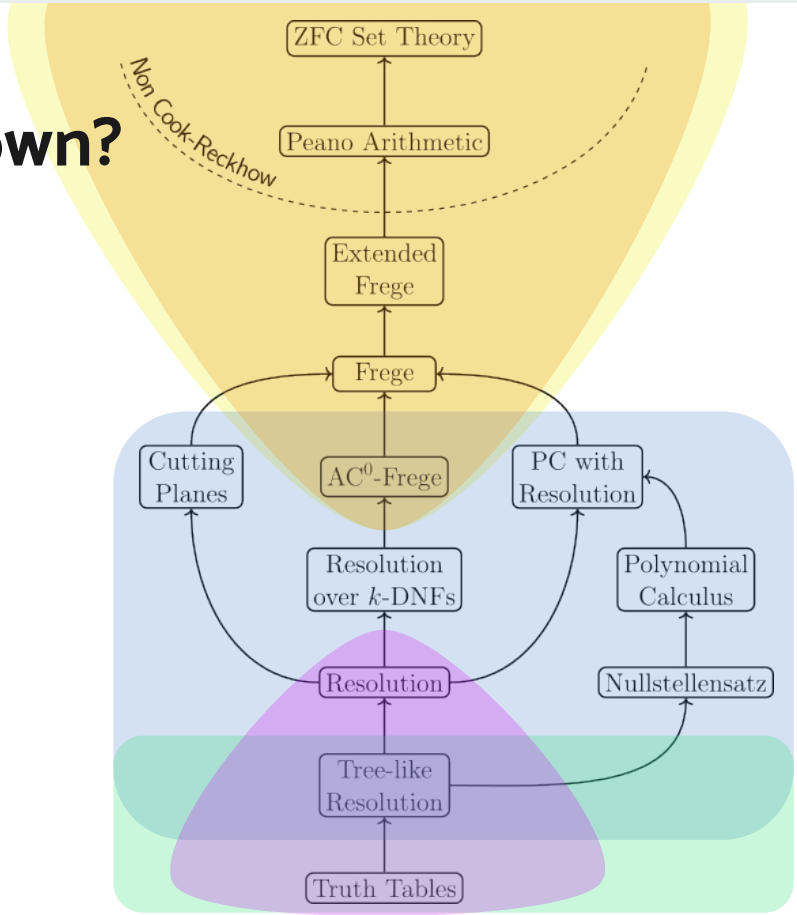
A third property: analyzability

Can we extract satisfying assignments from proofs of lower bounds?



exponential lower bound on $\mathrm{Ref}_s(\varphi)$ ⟷ Q analyzable

What is known?



Non-analyzable
Unless P = NP (or unless QP ⊆ NP)

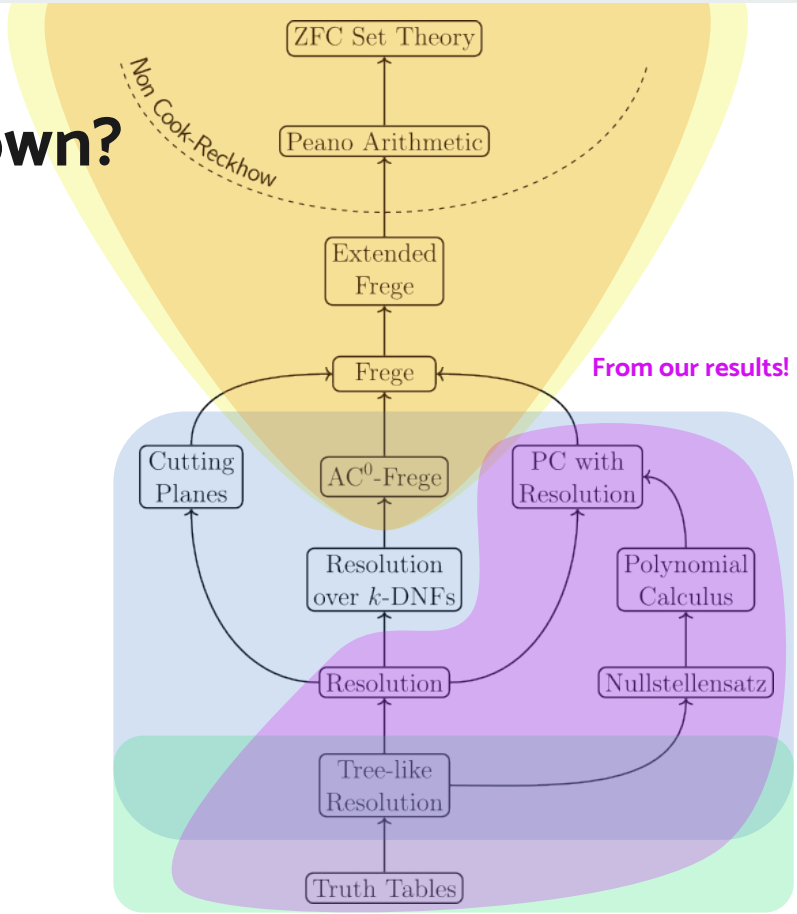
Non-automatable
Unless Diffie-Hellman fails

Non-automatable
Unless P = NP (or unless ETH fails)

(Quasi) Automatable!

Analyzable!

What is known?



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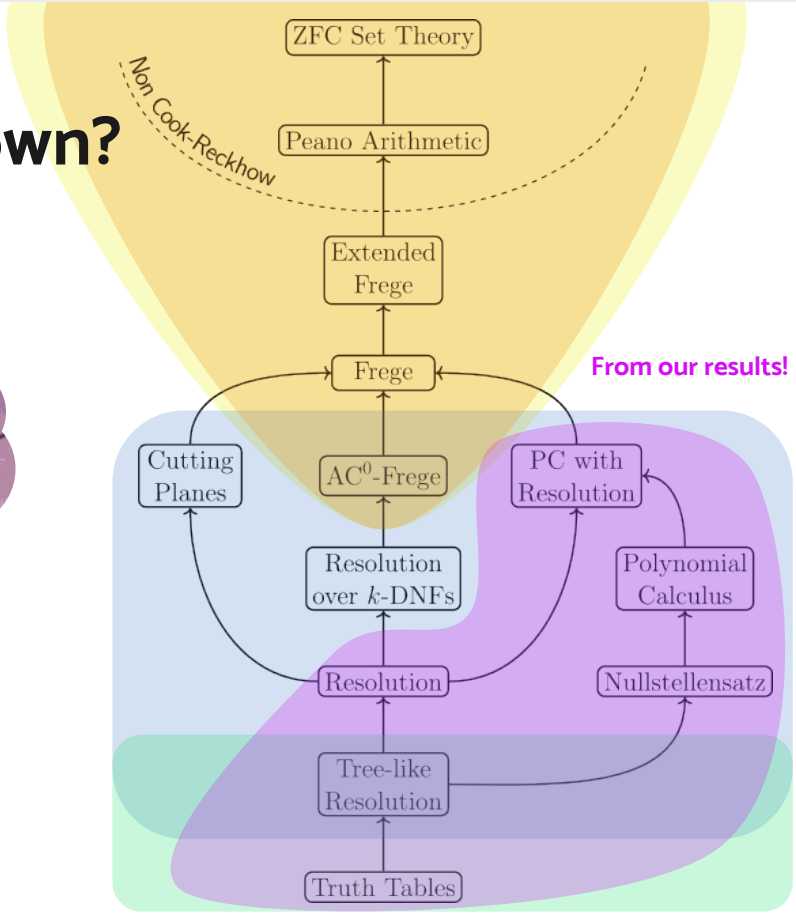
(Quasi) Automatable!

Analyzable!

What is known?



Looks suspicious...



Non-analyzable
Unless P = NP (or unless QP ⊆ NP)

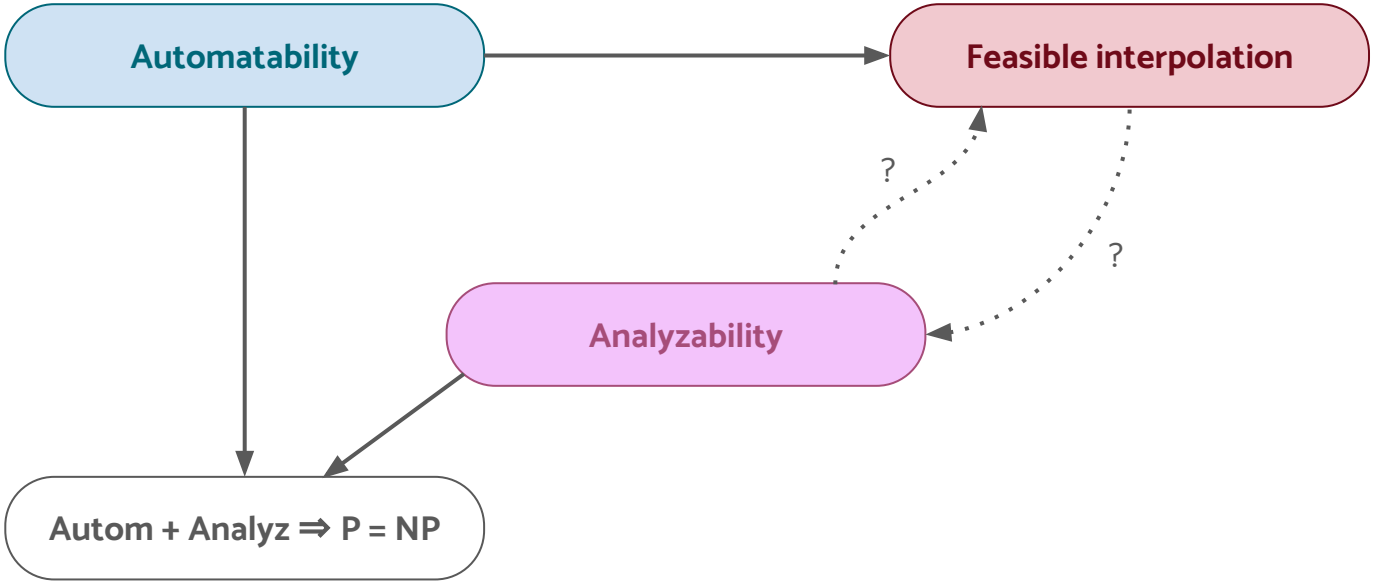
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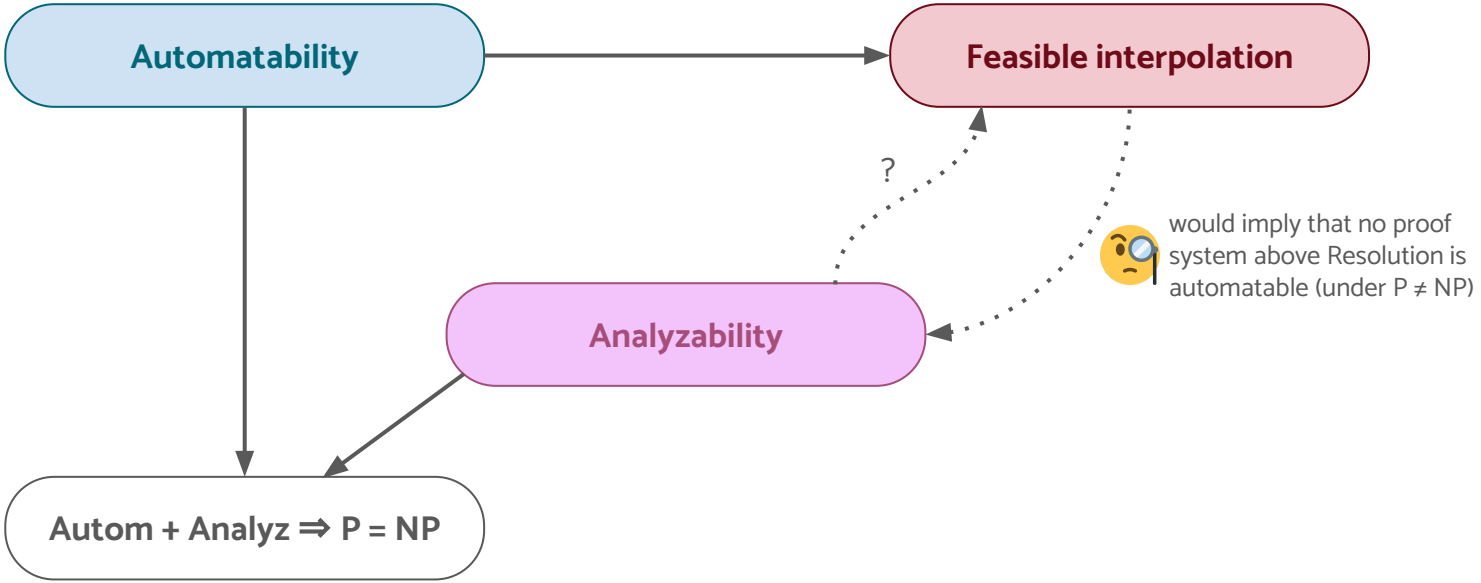
(Quasi) Automatable!

Analyzable!

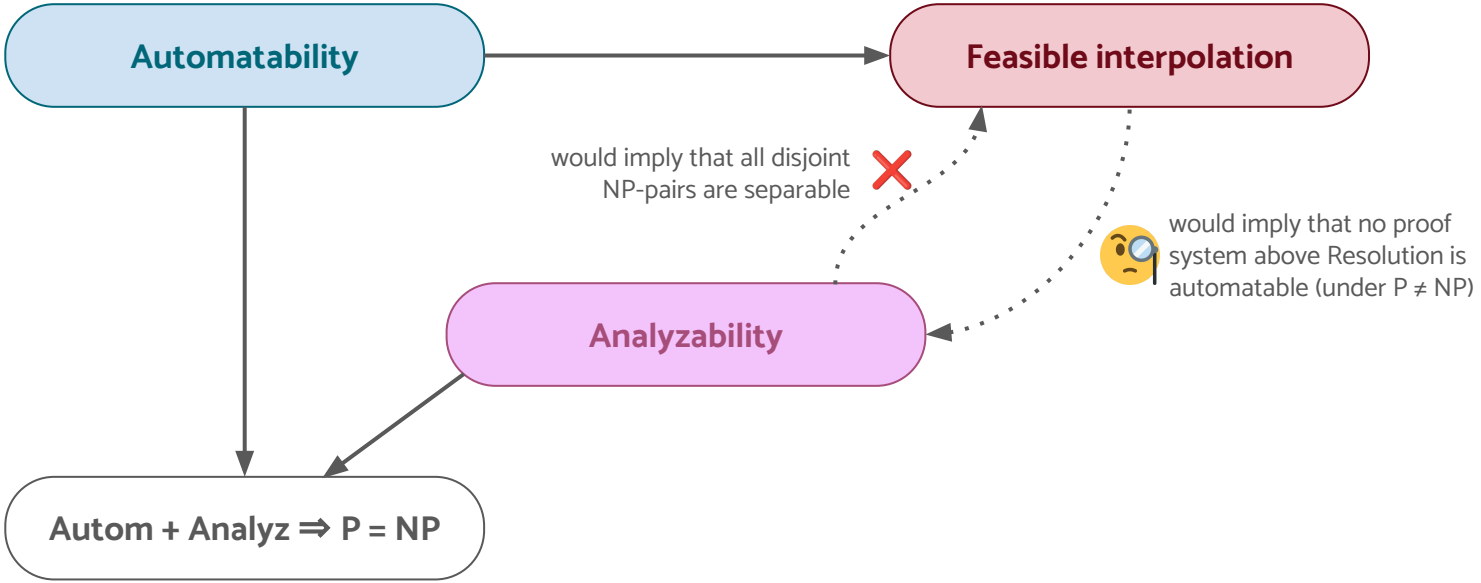
New investigation



New investigation



New investigation



Recap of contributions

✓ **Improvements for $\mathrm{Ref}_s(\varphi)$**

result now holds for any $s \geq n^2$, in Resolution and PCR

? **F.I. (likely) doesn't imply analyzability**

at least hard as proving that no proof system above Resolution is automatable (under $P \neq NP$)

✓ **Analyzability of algebraic systems**

all systems up to PCR are analyzable

⇏ **Analyzability doesn't imply F.I.**

unless all disjoint NP-pairs are separable

Open problems

1 **Analyzability in other systems**

are there other analyzable proof systems?

2 **Another route to analyzability**

can analyzability be proved in a simpler way?

3 **Automatability vs. Analyzability**

we know that analyzable ⇒ NP-hard to automate, what about the converse?

4 **An unconditional separation**

can analyzability and feasible interpolation be unconditionally separated?

Thank you for your attention!

Questions?